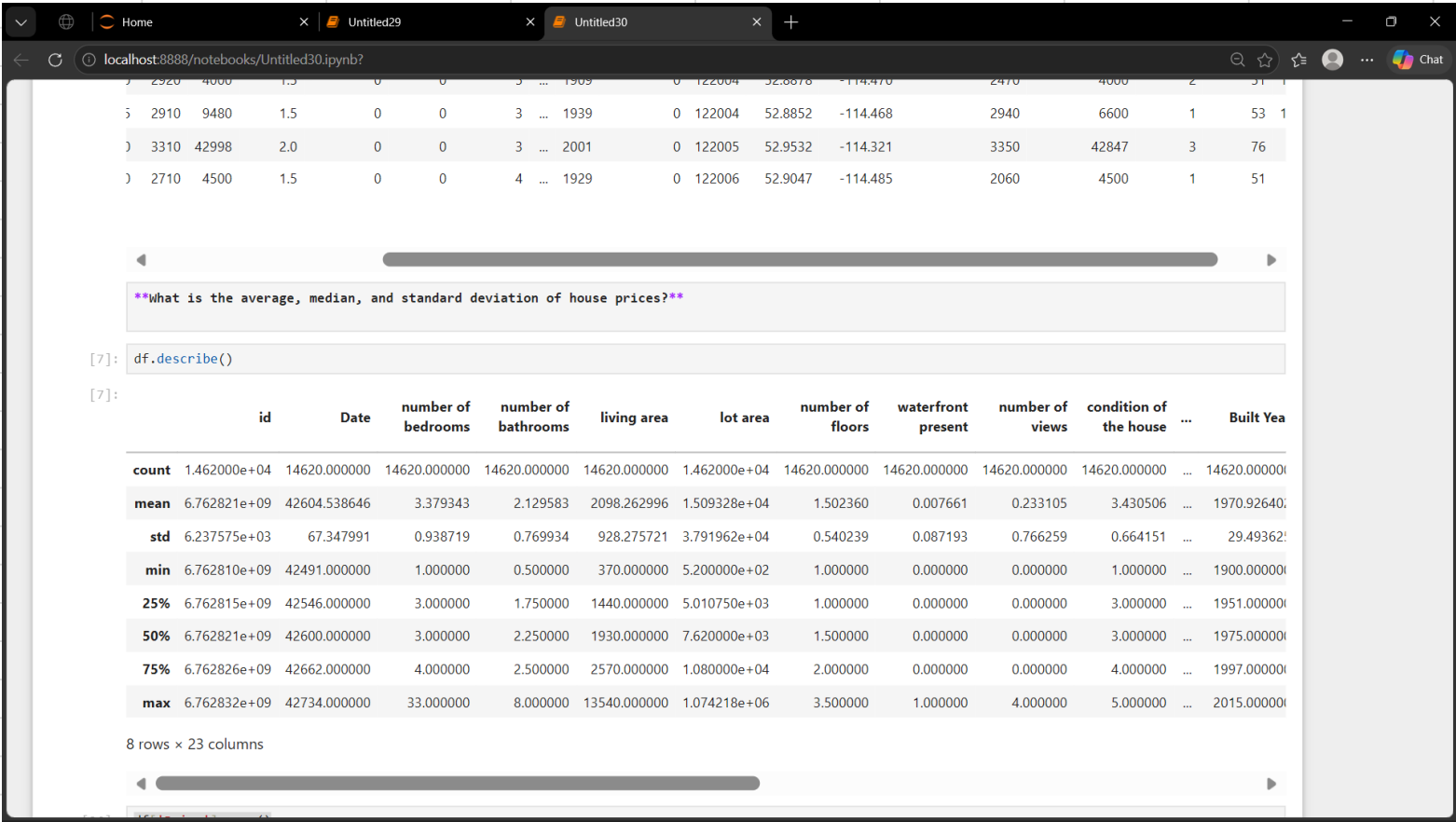


Exploratory Data Analysis (EDA Basics)

QUESTION 1

What is the average, median, and standard deviation of house prices?

ANSWER



QUESTION 2

Which number of bedrooms is most common?

ANSWER

```
[28]: df['Price'].std()

[28]: 367532.3808039669

**Which number of bedrooms is most common?**

[31]: df['number of bedrooms'].value_counts()

[31]: number of bedrooms
3      6612
4      4724
2      1844
5      1079
6       176
1       136
7         30
8         11
9          3
10         3
33         1
11         1
Name: count, dtype: int64
```

QUESTION 3

Check if the price data is skewed.

ANSWER

```
3      6612
4      4724
2      1844
5      1079
6       176
1       136
7        30
8         11
9          3
10         3
33         1
11         1
Name: count, dtype: int64

**Check if the price data is skewed.**

[32]: df['Price'].skew()

[32]: np.float64(4.269297720707117)

[ ]:
[ ]:
[ ]:
[ ]:
```

Interpretation:
Skewness = 0 → Symmetric distribution
Skewness > 0 → Right-skewed (long tail on the right)
Skewness < 0 → Left-skewed (long tail on the left)

QUESTION 4

What is the average price for each number of bedrooms?

ANSWER

```
Home x Untitled29 x Untitled30 x +
localhost:8888/notebooks/Untitled30.ipynb
[32]: np.float64(4.20929720707117)

**What is the average price for each number of bedrooms?**

[36]: df.groupby('number of bedrooms')['Price'].mean()

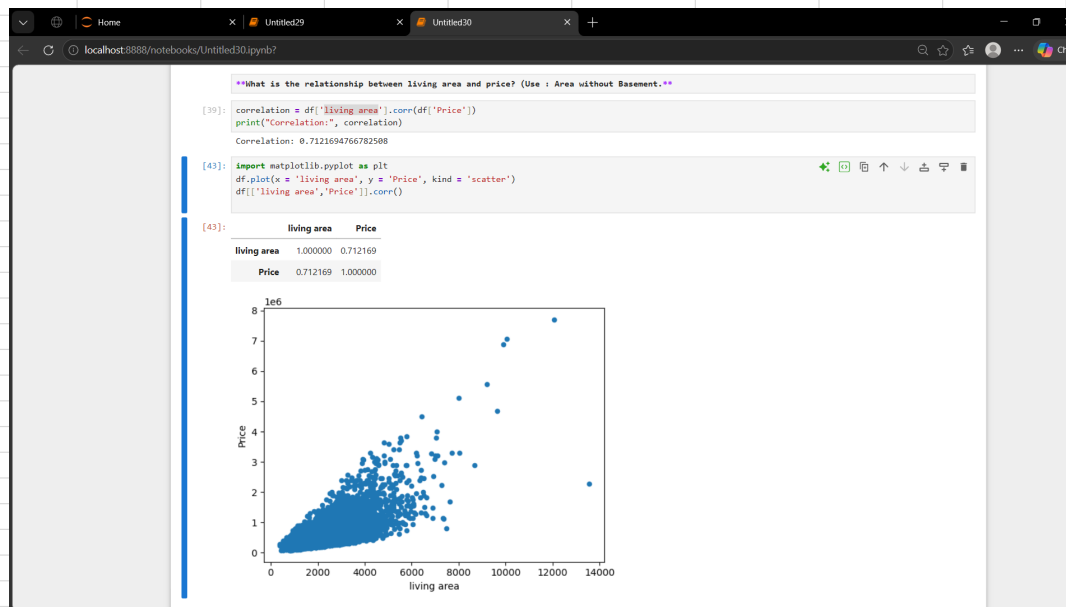
[36]: number of bedrooms
1      3.089638e+05
2      3.985476e+05
3      4.632776e+05
4      6.361988e+05
5      7.752550e+05
6      8.375815e+05
7      1.016544e+06
8      1.208455e+06
9      7.766663e+05
10     8.200000e+05
11     5.200000e+05
33     6.400000e+05
Name: Price, dtype: float64

[ ]:
```

QUESTION 5

What is the relationship between living area and price? (Use : Area without Basement).

ANSWER



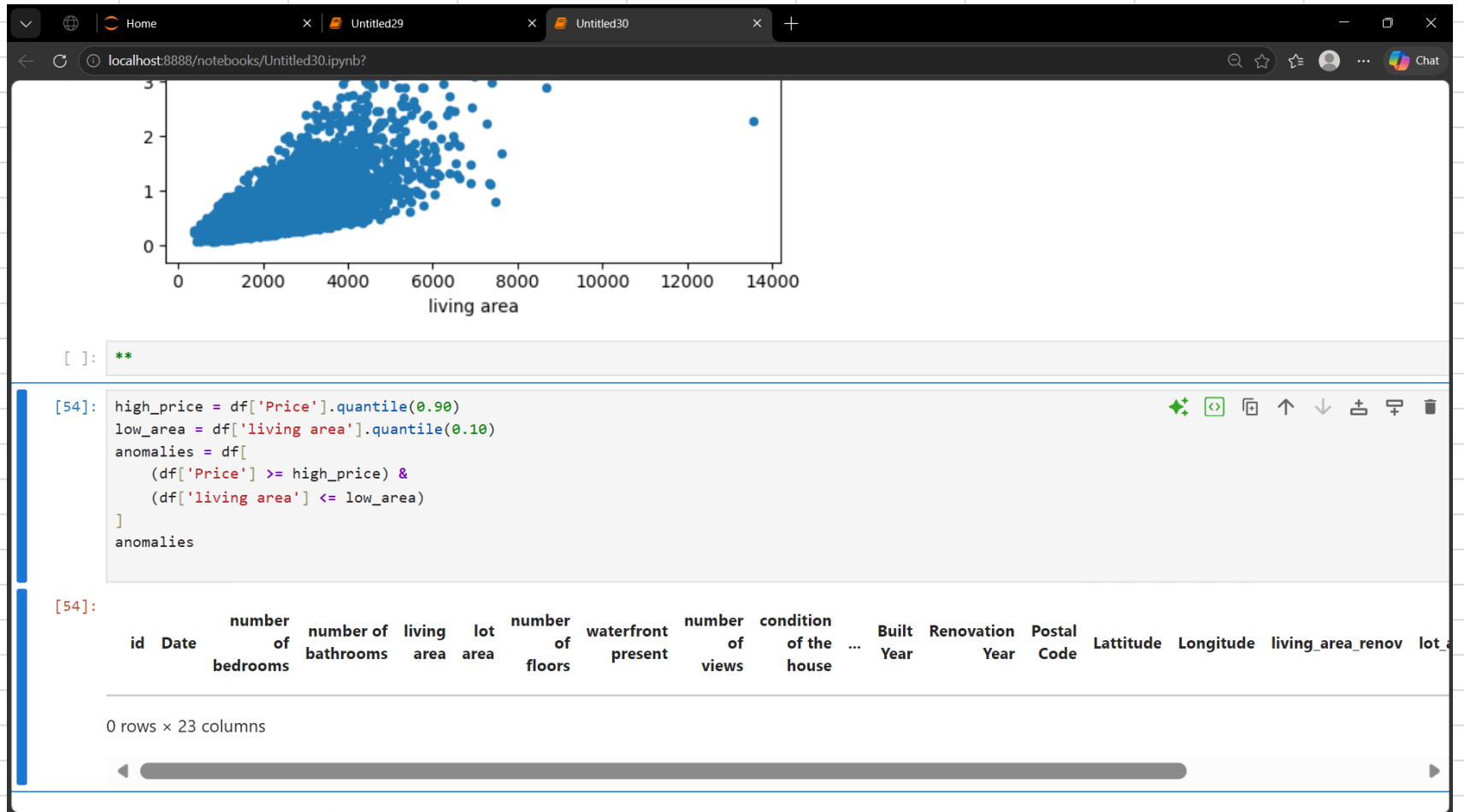
conclusion

If the correlation is positive and the scatter plot shows an upward trend
There is a positive relationship between living area and price.

QUESTION 6

Identify any anomalies where houses have high prices but low areas.

ANSWER



QUESTION 7

Compare average price based on number of floors and houses with or without waterfront.

ANSWER

```
0 rows × 25 columns

**Compare average price based on number of floors and houses with or without waterfront.**

[57]: df.groupby('number of floors')['Price'].mean()

[57]: number of floors
1.0    4.369769e+05
1.5    5.549301e+05
2.0    6.487372e+05
2.5    1.186233e+06
3.0    5.663281e+05
3.5    1.205875e+06
Name: Price, dtype: float64

[59]: df.groupby('waterfront present')['Price'].mean()

[59]: waterfront present
0    5.304174e+05
1    1.641902e+06
Name: Price, dtype: float64

[60]: df.groupby(['number of floors', 'waterfront present'])['Price'].mean().reset_index()

[60]:
```

	number of floors	waterfront present	Price
0	1.0	0	4.323724e+05
1	1.0	1	1.211081e+06
2	1.5	0	5.476162e+05
3	1.5	1	1.232507e+06
4	2.0	0	6.357742e+05
5	2.0	1	2.075931e+06
6	2.5	0	1.087654e+06
7	2.5	1	3.280000e+06
8	3.0	0	5.558095e+05
9	3.0	1	1.655000e+06
10	3.5	0	1.205875e+06

QUESTION 8

Identify the minimum and maximum house price. What does this indicate?

ANSWER

The screenshot shows a Jupyter Notebook interface with a browser window at localhost:8888. The notebook contains a table with 4 rows and 4 columns, followed by a code cell and its output.

7	2.5	1	3.280000e+06
8	3.0	0	5.558095e+05
9	3.0	1	1.655000e+06
10	3.5	0	1.205875e+06

****Identify the minimum and maximum house price. What does this indicate?****

```
[61]: df['Price'].agg(['min', 'max'])
```

```
[61]: min      78000
      max    770000
      Name: Price, dtype: int64
```

Price Range

The difference between the minimum and maximum price shows how widely house prices vary in the dataset.

Potential Outliers

If the maximum price is much larger than most other prices, there may be luxury properties or outliers.

QUESTION 9

Which location (zipcode/area) has the highest average price?

ANSWER

****Which location (zipcode/area) has the highest average price?****



```
[63]: avg_price_zip = df.groupby('Postal Code')['Price'].mean()  
      avg_price_zip.sort_values(ascending=False).head(1)
```

```
[63]: Postal Code  
      122071    2.348311e+06  
      Name: Price, dtype: float64
```

QUESTION 10	Write at least 5 insights from your analysis.											
ANSWER	1) The most common bedroom count in the dataset is 3 bedrooms, indicating that medium-sized family homes dominate the housing market.											
	2)There is a positive relationship between living area and house price.											
	3)Houses with a waterfront view have a much higher average price than houses without waterfront access. This suggests that location and scenic views add substantial value to a property.											
	4)Properties with more floors generally have higher average prices.											
	5) A few houses have very high prices despite relatively small living areas.											